

for the same has only been tested on Windows and Linux.

User's point of view

Every user has his or her experience and story to convey with different versions of the same software. This tool has mixed reviews; while some people like GHDL so much they want to use it again, others have faced plenty of issues with one particular version and they do not want to try other newer versions.

One such user at <http://sourceforge.net/> finds GHDL to be outstanding with Xilinx's Unisim and Xilinx-CoreLib packages without any issue. He appreciates the development team for building such a tool, while another user faces some issues with GHDL, when used with Xilinx's Unisim.

You could find some users at <http://www.sigasi.com/> who find it hard to compile and distribute GHDL at Linux platform; however, some find it slow. Some users also realise that a designer is never interested in compiling the compiler but would want to compile codes he or she generates. They also realise that GHDL is for small circuits and FPGAs that do not need much high speed, as in the cases of SoCs and ASICs.

There are some users at <http://www.edaboard.com/> who have had issues compiling and building the GHDL source code for version 0.29 but they also found a way around for that, which worked fine later.

GHDL aims to implement all versions of VHDL that support all standards given by IEEE, although there is still a long way to go before this can be achieved.

To sum up, GHDL can be a good compilation and simulation tool to

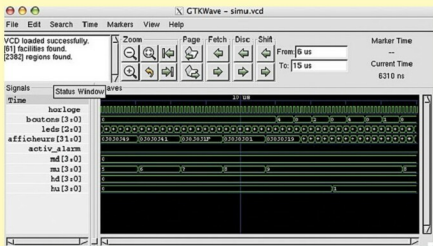


Fig. 2: GTKWave simulation

How to install

It is very easy to install GHDL on Windows if it is a pre-built version. You only need to open (or double-click) the GHDL installer. However, the Linux version of the tool is a command-line or binary version and needs some attention.

1. Unless links are set, the default location for binary installation cannot be altered.
2. Installation will take place in the following directory.
(usr/local directory (in a Linux system))
3. You should be at the root directory to install this.
4. Use the following commands for installation:

```
$ su
# tar -C -j -xvf @TARFILE@.tar.bz2
```

The mcode version of this tool can also be used for installation on Windows (32-bit) and Mac OS X version 10.5/10.6/10.7/10.8, i386 (32-bit). But, it is important that you have administrative privileges. Following are the steps for installing the mcode version:

On Windows:

1. Create an installation directory as needed, for example, C:\ghdl
2. Unzip ghdl-0.31-mcode-win32.zip into C:\ghdl
3. Open a command shell and cd into the resulting directory, C:\ghdl\ghdl-0.31-mcode-win32
4. Run the following batch files:
`set_ghdl_path.bat`
`reanalyze Libraries.bat`
5. Check the installation

use with small circuits and FPGAs, especially when it costs almost nothing. We understand that it would need a few other graphical and support tools, but this also means that other workable open source or free tools can be easily used with it to

make it more useful.

A simulator, compiler and writer, GHDL plays all these roles together. A copy of GHDL is included in this month's DVD for EFY Plus. ●

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